母粒子静止

$$\boldsymbol{p}_n = 0$$

$$\gamma: \ E_\gamma = h\nu$$

$$C = p + e + \bar{\nu}_e$$

$$\boldsymbol{p}_C = -\boldsymbol{p}_\gamma$$

$$m_n c^2 = E_\gamma + E_C$$

母粒子静止系

整体反冲不为零

变换到

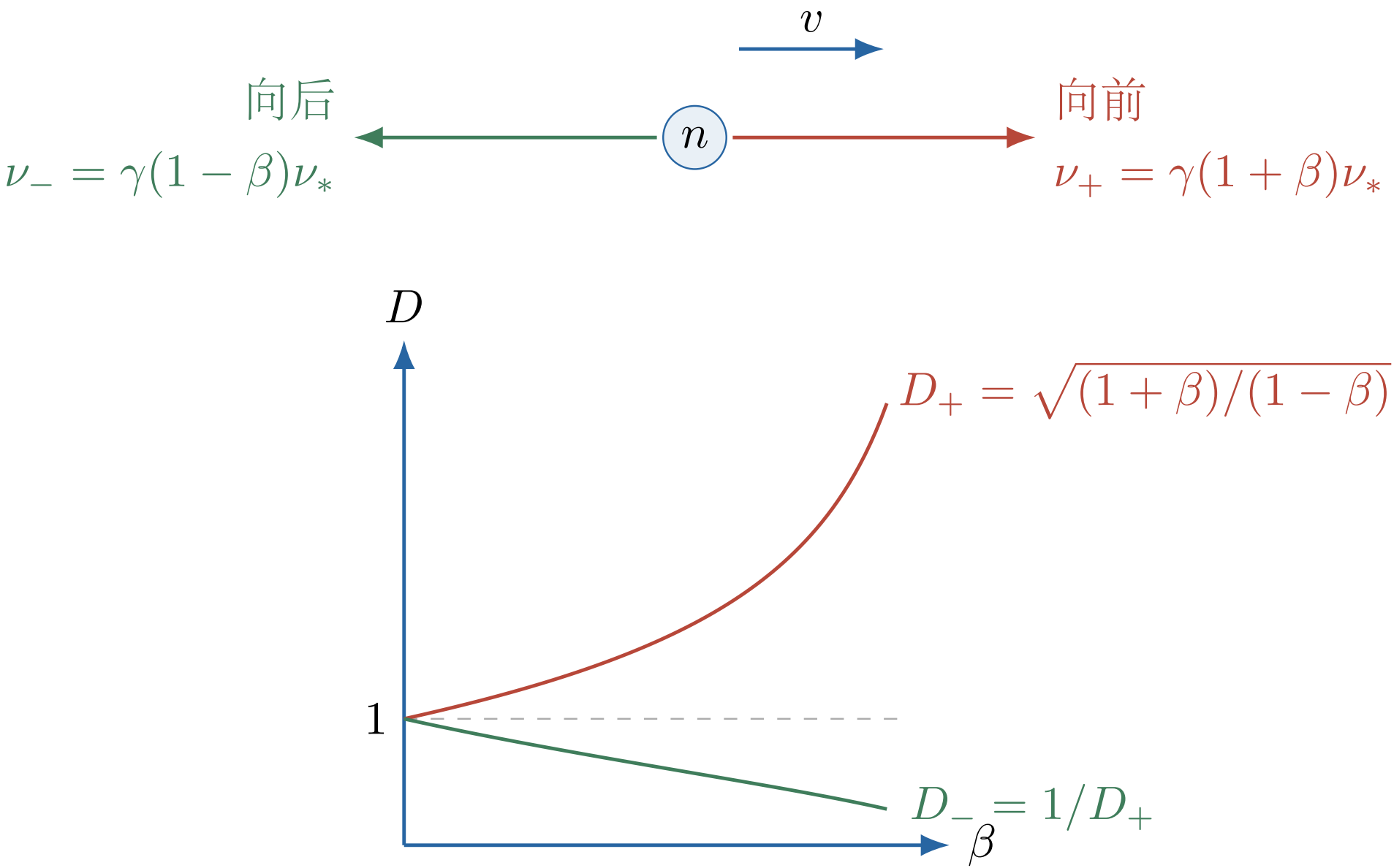
C 的质心系

端点极限

内部能量最小

$$E_C^2 - p_C^2 c^2 = M_C^2 c^4$$

$$M_C \geq m_p + m_e$$


v
向后
$\nu_- = \gamma(1-\beta)\nu_*$
n
向前
$\nu_+ = \gamma(1+\beta)\nu_*$
D
$D_+ = \sqrt{(1+\beta)/(1-\beta)}$
1
$D_- = 1/D_+$
β